

CODING TEMPLE
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Managing BYOD and Mobile Device Security Using Microsoft Intune (MDM & MAM)



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Executive Summary

This report documents the configuration and validation of a Bring Your Own Device (BYOD) program using Microsoft Intune (MDM & MAM) and Microsoft Entra Conditional Access.

- **Goal:** Protect corporate data while respecting employee privacy.
- **Scope:** Windows BYOD (MDM + Compliance + CA), iOS BYOD (MAM + CA).
- **Outcome:** Compliant Windows devices and managed mobile apps access resources; non-compliant endpoints and unapproved data flows are blocked.

1) BYOD Security Policy

Scope:

Applies to employees/contractors using personal devices (Windows/macOS/iOS/Android) to access company data.

Acceptable Use: BYOD used only for approved business tasks with approved apps.

Enrollment & Access:

- Windows/macOS: MDM enrollment (Intune) + device must be **Compliant**.
- iOS/Android: MAM (App Protection) on corporate apps (no full device control).

Minimum Security:

- Full-disk encryption (BitLocker/FileVault/native).

- Strong password/PIN, auto-lock, lockout escalation.

- No **rooted/jailbroken** devices.

Data Protection (MAM/DLP):

- Access only via managed apps (Outlook, Teams, OneDrive, Word, Excel, PowerPoint).

- Restrict **copy/paste**, **open-in/share**, and **Save As** to corporate storage (OneDrive/SharePoint).

- Links open in **Microsoft Edge** (mobile).

Authentication: MFA required for cloud apps.

Monitoring & Privacy: IT manages only corporate data/apps; personal content remains private.

Incident Response: Report loss/theft within 24h; selective wipe may be performed.

Acknowledgment: Users must accept policy before access.

Microsoft Intune admin center

Home > Apps > Protection >

Create policy

PIN for access

PIN type: Required (selected) / Not required

Simple PIN: Allowed (selected) / Blocked

Select minimum PIN length: 8

Touch ID instead of PIN for access (iOS 8+ / Windows Hello): Allowed (selected) / Blocked

Override biometrics with PIN after timeout: Required (selected) / Not required

Timeout (minutes of inactivity): 30

Face ID instead of PIN for access (iOS 11+ / Windows Hello): Allowed (selected) / Blocked

PIN reset after number of days: No (selected) / Yes

Number of days: 30

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Microsoft Intune admin center

Home > Apps > Protection >

Create policy

Basics Apps Data protection Access requirements **Conditional launch** Assignments Review + create

Set the sign-in security requirements for your access protection policy. Select a **Setting** and enter the **Value** that users must meet to sign in to your company app. Then select the **Action** you want to take if users do not meet your requirements. In some cases, multiple actions can be configured for a single setting. [Learn more about conditional launch actions.](#)

Setting	Value	Action
Max PIN attempts	5	Reset PIN
Offline grace period	1440	Block access (infinite)
Offline grace period	80	Wipe data (play)

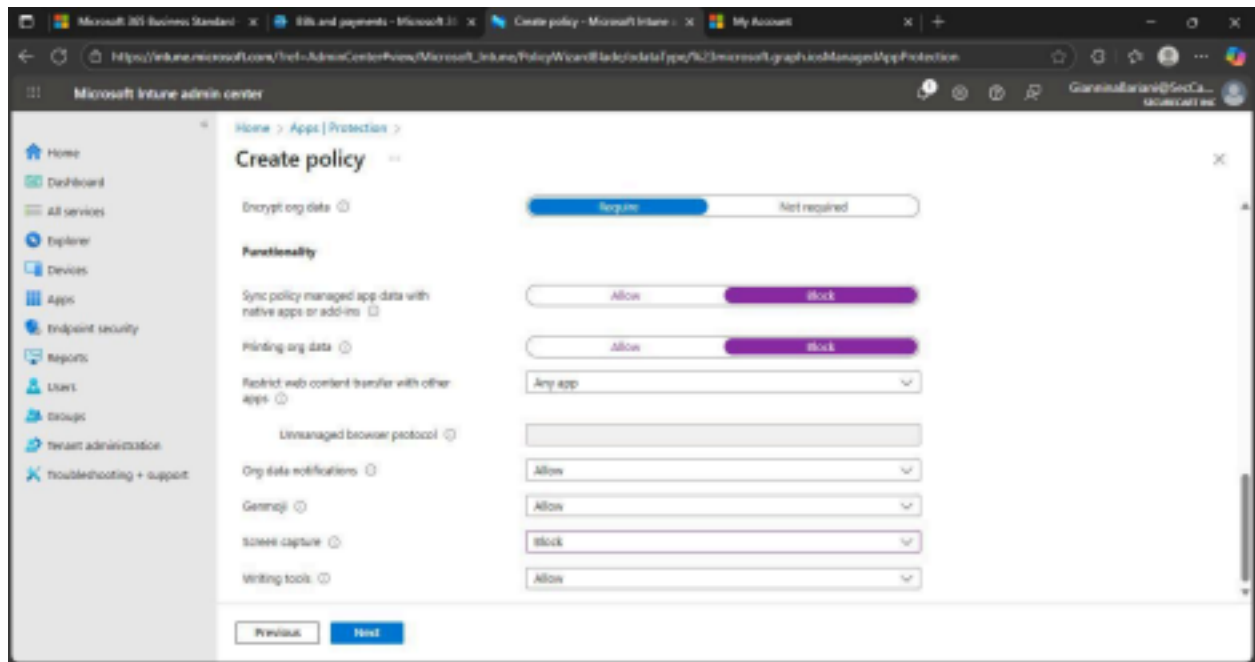
Select one

Device conditions

Configure the following conditional launch settings for device-based conditions through your app protection policy.

[For the device-based conditions, see the conditional launch actions. Learn more about conditional launch actions.](#)

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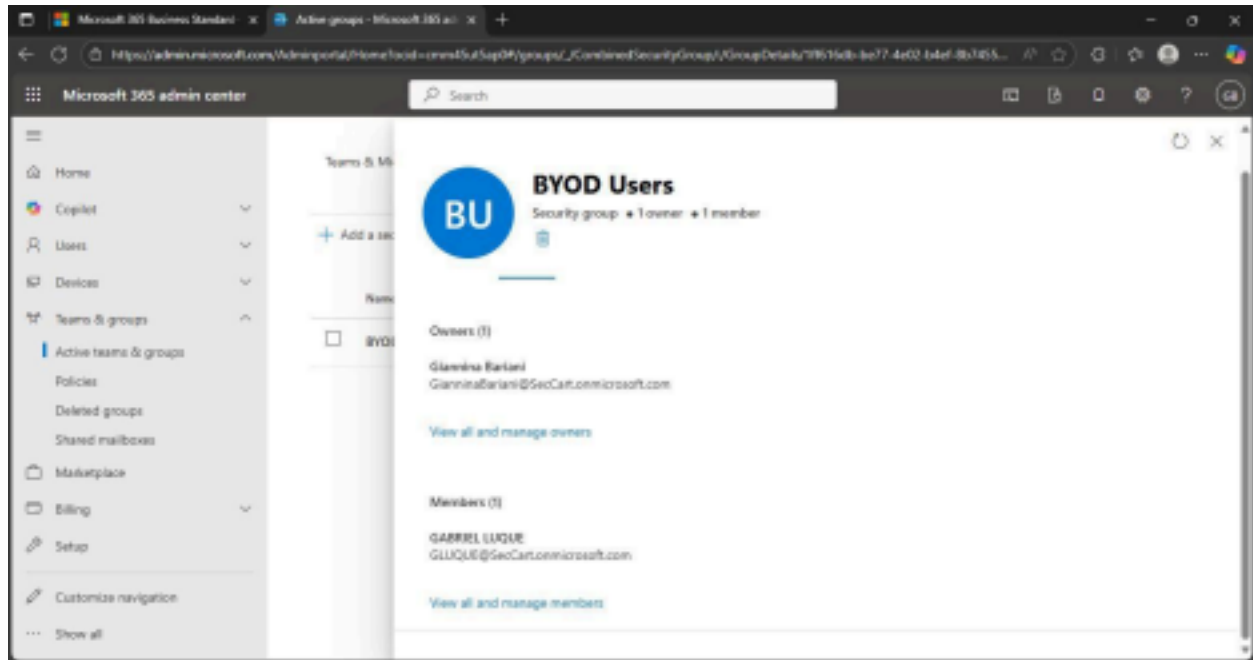
2)

Environment & Accounts

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- **Tenant:** SecureCart Inc (lab).
- **Primary user:** GLUQUE@SecCart.onmicrosoft.com (member of **BYOD Users** group).
- **BYOD group:** **BYOD Users** (scope for assignments).
- **Test devices:**

- Windows 11 personal laptop (**RETRONIKAA**) — MDM/Compliance.
- iPhone — MAM (managed apps only).



3) Intune Configuration

3.1 Device Enrollment (Windows BYOD)

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- **Path:** Windows **Settings** → **Accounts** → **Access work or school** → **Connect** → sign in → approve MFA → **Info** → **Sync**.
- Device is shown in **Intune** → **Devices** → **All devices** and marked **Compliant** once policies apply.





3.2 Compliance Policy (MDM) — Windows 10/11

Policy: **BYOD WINDOWS** (assigned to **BYOD Users**).

Key settings:

- **Encryption:** BitLocker required; **TPM**, **Secure Boot**, **Code Integrity** required. •

Password: Minimum length/complexity; inactivity lock; block simple passwords.

- **Security stack:** Defender Antivirus/Anti-spyware/Firewall required.

Actions: Mark non-compliant immediately; send email notification.





3.3

App Protection Policies (MAM) — iOS (and Android if applicable)

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Policy (iOS): [BYOD iOS – Data Protection](#) (assigned to **BYOD Users**).

Targeted apps: Outlook, Teams, OneDrive, Word, Excel, PowerPoint.

Data protection:

- **Send/Receive org data: Policy-managed apps** (with Open-In/Share filtering).
- **Cut/Copy/Paste: Policy-managed apps (paste-in only).**
- **Save copies of org data: Block** (or **OneDrive/SharePoint only**).
- **Encrypt org data: Required.**
- **Restrict web content to: Microsoft Edge.**
 - Access requirements:** Require PIN (≥6) + biometric; periodic recheck.
 - Conditional launch: Block** jailbroken/rooted.







3.4 Conditional Access (Microsoft Entra)

Policy A (Windows/macOS): BYOD – Win/macOS – Require compliant + MFA

- **Users:** BYOD Users
- **Apps:** Microsoft 365 (Office 365)
- **Device platforms:** Windows, macOS
- **Grant:** Require **MFA** and **device marked as compliant** (Require all)

- **State: On** (after verifying in Report-only)

Policy B (Mobile): BYOD – iOS/Android – Require App Protection + MFA

- **Users:** BYOD Users
- **Apps:** Microsoft 365 (Office 365)
- **Device platforms:** iOS, Android
- **Grant:** Require **MFA** and **App protection policy**
- **State: On** (after verifying in Report-only)



4) Validation & Test Results

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4.1 Windows (MDM + CA)

- **Allowed (Compliant):** From the compliant Windows device, sign in to **Outlook/Teams**

→ access succeeds; MFA enforced.

Denied (Non-compliant): Temporarily increase **Minimum password length** in the compliance policy to exceed current password → **Sync** device → status becomes **Non-compliant**. Attempt to access Outlook/Teams → **blocked by CA**. Revert policy, **Sync**, device back to **Compliant**.

4.2 iOS (MAM)

- **Screenshot blocked** inside Outlook (verified).
- **Copy/Paste to personal app (Notes)** blocked.
- **Save/Share** to personal storage/apps blocked; **OneDrive** allowed.

Result: Data remains inside managed apps/storage; personal OS/apps remain private.

5) Privacy Considerations

- **Mobile BYOD (MAM):** IT manages only corporate apps/data (Outlook/Teams/OneDrive/Office). Personal photos, messages, and personal apps are **not** visible to IT.

- **Windows BYOD (MDM):** Only security posture and compliance are enforced (encryption, password, Defender, Secure Boot, etc.); personal files remain private.
- **Selective wipe** removes only corporate data from personal devices.

6) Audit & Review

Monthly:

- Intune **Device compliance** report (top reasons for non-compliance).
- **App protection** status (coverage for target users).
- **Conditional Access** sign-in logs (denied/required controls).

Quarterly: Random user sample to confirm BYOD acknowledgment and correct enrollment.

Annually: Full policy review by Security & IT; update based on threat/tech/regulatory changes.

7) Conclusion

TechSecure Innovations' BYOD program leverages Intune MDM/MAM and Entra Conditional Access to enforce encryption and strong identity, keep corporate data inside managed apps and storage, and prevent exfiltration to personal apps — all while preserving user privacy. Validation proves compliant devices are allowed and policy-breaking scenarios are blocked.

Appendix A — Employee BYOD Acknowledgment (Template)

I acknowledge that I have read and agree to comply with TechSecure Innovations' BYOD Security Policy. I understand that corporate data will be accessed only via approved, managed applications; that my device must meet security requirements (encryption, screen lock, updates); and that the company may selectively wipe corporate data from my device if it is lost, stolen, or upon my separation from the company.

Name: _____ **Dept:** _____ **Date:** _____

Signature: _____

